AI-Powered Multimodal Misinformation Detection and Verification System

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October 2026



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Introduction

- Misinformation spreads rapidly across digital platforms through misleading text, manipulated images, and AI-generated content, making reliable detection increasingly important.
- This project develops an AI-powered multimodal system using DistilBERT for text analysis and EfficientNet-B0 for image analysis, combining their features to classify content as real or fake.



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Problem Statement

The lack of reliable multimodal systems for detecting misinformation across text and images makes identifying misleading digital content challenging.



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Literature Review

Paper Title & Year	Methodology Used	Key Gaps Identified
A Multi-Level Fusion-Based Framework for Multimodal Fake News Classification Using Semantic Feature Extraction (2025)	Combines CNN-based visual features, RNN-based textual features, multiple fusion strategies, and polynomial-kernel classification.	Performance varies substantially across datasets, indicating challenges in generalization across different data domains.
Advanced Machine Learning Techniques for Fake News (Online Disinformation) Detection: A Systematic Mapping Study (2021)	Systematic mapping study reviewing machine learning methods, datasets, and information sources used for disinformation detection.	Identifies broader research challenges in reliable detection, dataset selection, and evaluation across different approaches.
Multimodal Fake News Detection Through Intra-Modality Feature Aggregation and Inter-Modality Semantic Fusion (2024)	IFIS uses entity detection, attention-based feature aggregation, and two parallel co-attention blocks for text-image fusion.	The study identifies the need to incorporate additional modalities, such as video, audio, and propagation information.

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Paper Title & Year	Methodology Used	Key Gaps Identified
SSM: Stylometric and Semantic Similarity Oriented Multimodal Fake News Detection (2023)	Uses stylometric text features, headline-summary similarity, text-image semantic similarity, image forgery analysis, and feature fusion.	The multi-module pipeline increases system complexity; the approach primarily focuses on text and images.
SARD: Fake News Detection Based on CLIP Contrastive Learning and Multimodal Semantic Alignment (2024)	Uses CLIP-based contrastive learning to align textual and visual features with social-context information.	Depends on social-context features in addition to text and images, which may not always be available or reliable.
LLM-Enhanced Multimodal Detection of Fake News (2024)	FND-LLM combines text and visual branches, image-tampering analysis, co-attention, and an LLM reasoning branch.	Large language models increase computational requirements; conflicts between modalities and heterogeneous social-context data remain challenges.



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Proposed Solution

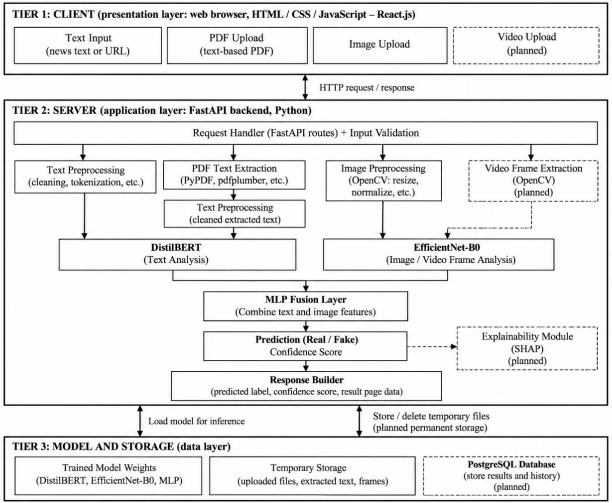
The proposed system detects misinformation using an AI-powered multimodal pipeline with the following components:

- **Text Analysis:** DistilBERT extracts contextual features from text to classify content as real or fake.
- **Image Analysis:** EfficientNet-B0 extracts visual features to identify potentially misleading image content.
- **Multimodal Feature Fusion:** An MLP combines text and image features to improve classification by using information from both modalities.
- **Web-Based Application:** A React.js frontend and FastAPI backend support text, text-based PDF, image, and text-image inputs, returning a predicted label and confidence score.

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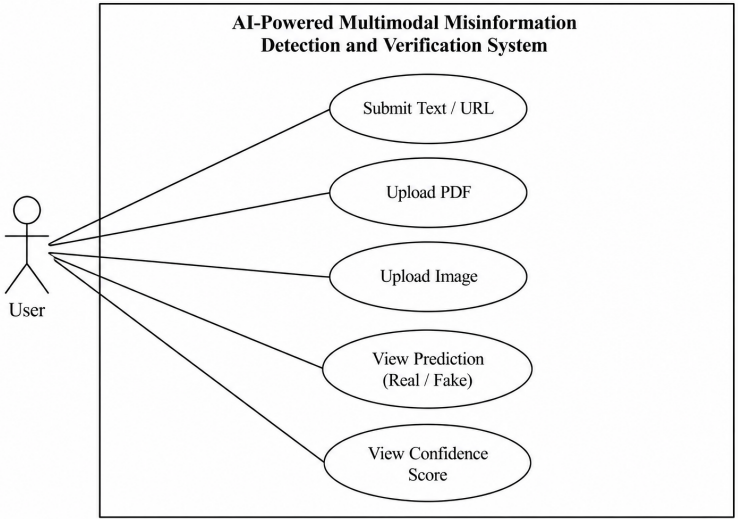
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Use Case Diagram

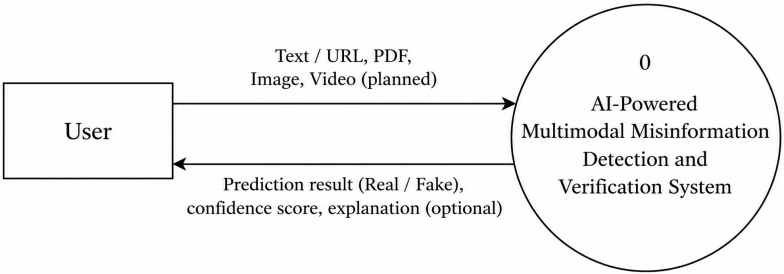
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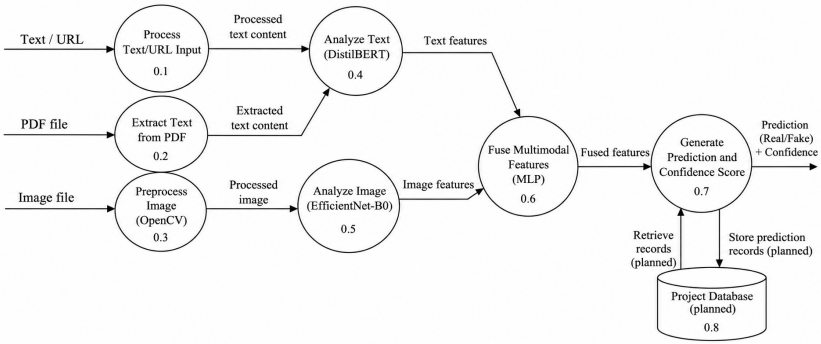
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Data Flow Diagram – Level 1





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Methodology

1. **Input Submission:** The user submits text, a URL, a text-based PDF, or an image through the React.js web interface.
2. **Input Processing:** The FastAPI backend validates the input and extracts text from PDF documents using PyMuPDF.
3. **Text Feature Extraction:** DistilBERT processes the text and extracts contextual features for misinformation classification.
4. **Image Preprocessing:** OpenCV preprocesses uploaded images and resizes them to the required model input dimensions.
5. **Image Feature Extraction:** EfficientNet-B0 extracts visual features from the preprocessed images.
6. **Multimodal Feature Fusion:** An MLP combines the extracted text and image features to support multimodal classification.



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Methodology

7. **Prediction:** The trained model classifies the supported input as Real or Fake.
8. **Confidence Estimation:** The system obtains a confidence score associated with the predicted class.
9. **Result Generation:** The FastAPI backend prepares the predicted label and confidence score for the frontend.
10. **Result Presentation:** The React.js interface displays the prediction and confidence score to the user.

Planned extensions: video analysis, OCR for scanned PDFs, SHAP explainability, and external fact verification.



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Functional Requirements

- **Input:** Accept text, URLs, text-based PDF documents, and images.
- **Preprocessing:** Extract text from PDFs and preprocess images for model input.
- **Text Analysis:** Use DistilBERT to extract textual features.
- **Image Analysis:** Use EfficientNet-B0 to extract visual features.
- **Classification:** Combine text and image features using an MLP for multimodal classification.
- **Result:** Display the predicted Real/Fake label and confidence score.



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Non-Functional Requirements

- **High Accuracy:** Provide reliable misinformation classification using text and image features.
- **Fast Performance:** Generate predictions within a reasonable response time.
- **Reliability:** Ensure consistent predictions and handle invalid inputs appropriately.
- **Scalability:** Support future increases in users, datasets, and input modalities.
- **Maintainability:** Allow easy updates to the models, backend, and frontend.

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- **Processor:** Intel Core i5 or AMD Ryzen 5 (or higher)
- **RAM:** 8 GB minimum; 16 GB recommended
- **Storage:** 256 GB SSD minimum; 512 GB recommended
- **Graphics:** NVIDIA GPU with CUDA support recommended for model training and faster inference



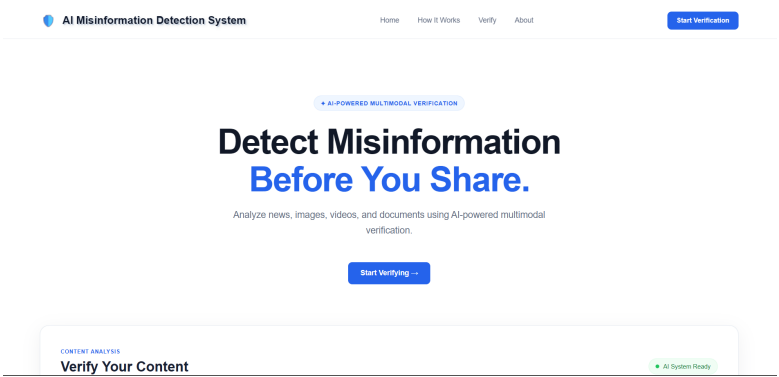
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Software Requirements

- **Programming Language:** Python 3.10+
- **Deep Learning Framework:** PyTorch
- **Text Analysis:** Hugging Face Transformers (DistilBERT)
- **Image Processing:** OpenCV and torchvision
- **Multimodal Fusion:** PyTorch MLP
- **Web Framework:** FastAPI
- **Frontend:** React.js, HTML, CSS, JavaScript
- **PDF Processing:** PyMuPDF

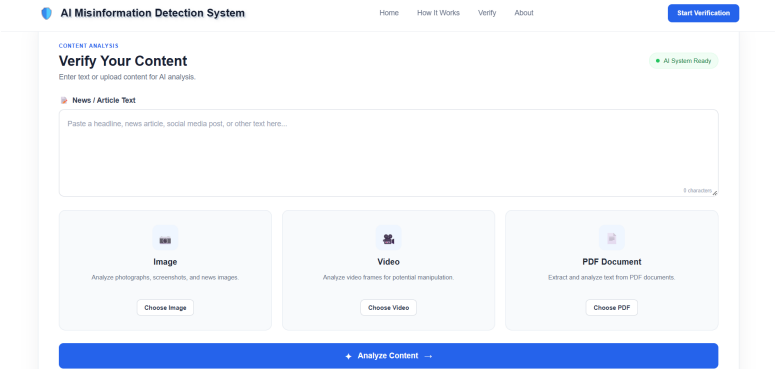
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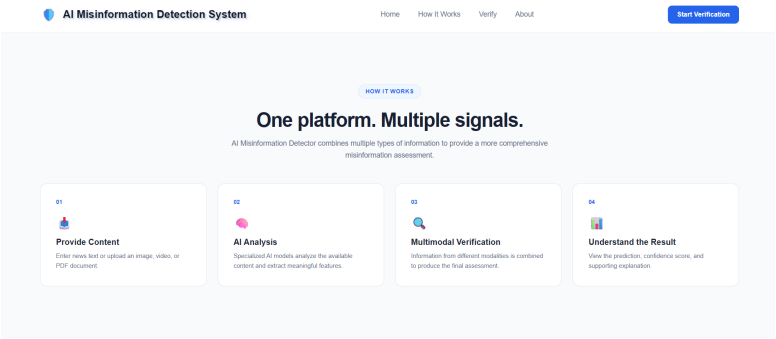
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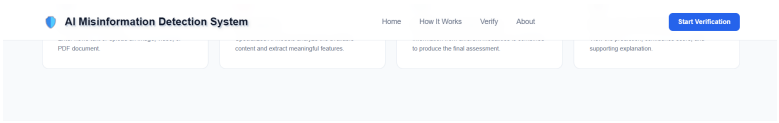
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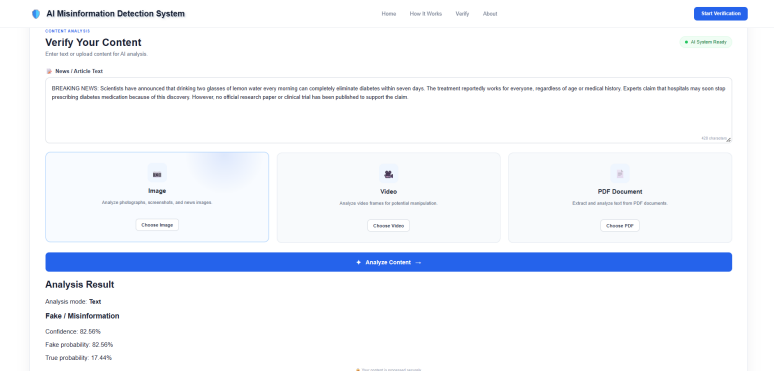
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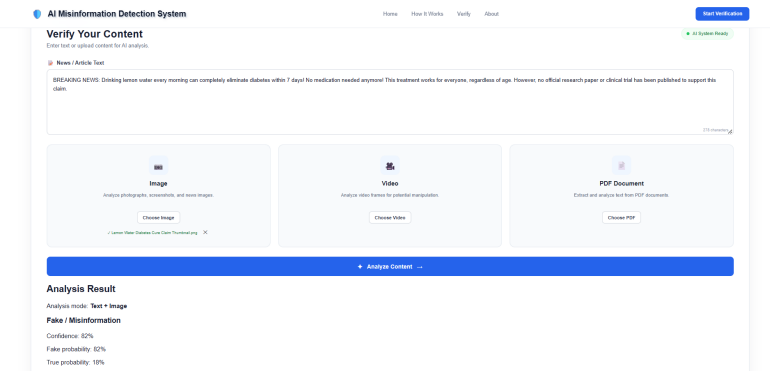
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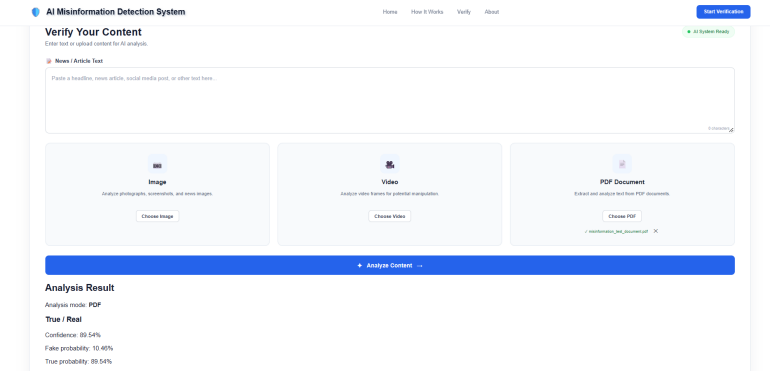
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Conclusion

- Develops an AI-powered multimodal system for detecting misinformation in digital content using text and images.
- Combines DistilBERT for textual feature extraction and EfficientNet-B0 for visual feature extraction.
- Uses an MLP-based feature fusion approach to combine text and image features for improved classification.
- Provides a web-based interface using React.js and FastAPI to analyse content and display Real/Fake predictions with confidence scores.



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- **Video and Audio Analysis:** Extending the system to analyse video frames and audio content to detect misinformation across multiple media formats.
- **External Fact Verification:** Integrating trusted news sources and reliable databases to verify claims against credible evidence.
- **Explainable AI:** Incorporating SHAP or other explainability techniques to highlight the features influencing model predictions.
- **Continuous Model Improvement:** Training and evaluating the models on diverse datasets to improve generalization, robustness, and detection performance.



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Literature Review Paper

Multimodal Fake News Detection: A Review of Machine Learning, Deep Learning, and Multimodal Fusion Approaches

Main Project Paper

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